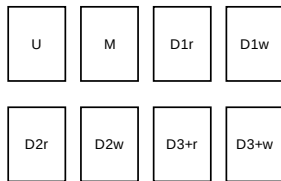


Pertussis transmission model schematic

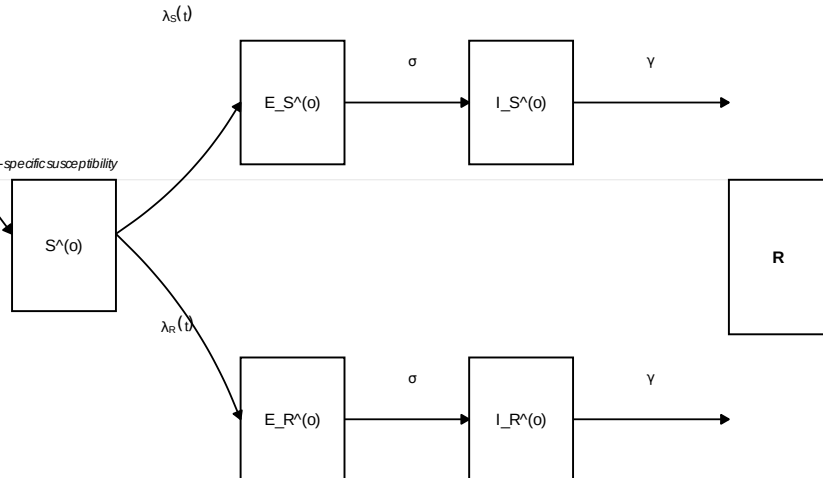
8 origin states



$\alpha = U, M, D1r, D1w, D2r, D2w, D3+r, D3+w$

Age omitted for clarity; the same template repeats across five age bands.

origin-specific susceptibility



The full ODE also retains treated states and origin history through infection.